

Yongjun Rong

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US Citizen

Summary

A product-minded Full-Stack Engineer with over 20 years of experience building and shipping scalable, user-centric applications. I excel at end-to-end feature development, from crafting responsive frontends with modern TypeScript frameworks to designing robust backend APIs and microservices with Python and Node.js. My recent focus has been on leveraging AI and real-time data to build compelling product experiences. I am passionate about clean code, software design patterns, and working collaboratively in fast-moving teams to turn ideas into products that users love.

Skills

- **Full-Stack Development:** Proven track record of building and shipping features across the entire stack, with strong skills in TypeScript/JavaScript, Python, modern frameworks, and REST API design.
- **Product-Centric Engineering:** A keen eye for user experience and a passion for building products that users love, with experience working closely with product managers and designers.
- **API & Microservice Design:** Expertise in designing clean, maintainable, and well-documented APIs and microservices using OpenAPI/Swagger, FastAPI, SpringBoot, and Node.js.
- **Real-Time & Data-Intensive Applications:** Experience building systems with real-time components, data streaming, and AI integration to create dynamic and responsive user experiences.
- **Software Design & Architecture:** Deep understanding of software design patterns, data structures, algorithms, and principles like SOLID and Clean Architecture to write scalable, maintainable code.
- **Cloud & DevOps Proficiency:** Strong familiarity with cloud platforms (AWS) and infrastructure-as-code (Terraform) to build, deploy, and scale applications effectively.
- **Collaboration & Adaptability:** Excellent team player with strong communication skills, a growth mindset, and the ability to quickly learn and apply new technologies in a remote setting.

Work Experience

Director of Cloud Architecture, Biophy

Feb., 2025 – Current, Cherry Hill, NJ (WFH), USA

As a hands-on director, I architected and implemented the full-stack infrastructure and data systems for an AI-powered RAG platform. I led the development of the data ingestion and API layers using Python, Django, and Pydantic, ensuring a clean, maintainable architecture while directly contributing to feature development.

- **Project: AI-Powered Data Ingestion and Scraping System for RAG**
 - Architected and delivered a full-stack data ingestion and scraping platform, applying SOLID and Clean Architecture principles to ensure a scalable, maintainable, and high-quality API and data processing layer.
 - Implemented FastAPI services to expose data transformation and ingestion as REST APIs.
 - Architected a modular data ingestion pipeline supporting multiple data sources and formats.
 - Designed data transformation layers following Clean Architecture principles.
 - Built robust error handling and retry mechanisms for data processing failures.
- **Technologies:** Python/Django/FastAPI/Pydantic, TypeScript/JavaScript, openAI API/LangChain, AWS/Lambda/SQS/SNS/EventBridge, Infrastructure as Code (Terraform/Terragrunt), SOLID Principles, Clean Architecture, REST APIs, Microservices

Senior Cloud Engineer II (Associate), McKinsey & Company

Dec., 2021 – Feb., 2025, Cherry Hill, NJ (WFH), USA

Led the full-stack development of genAI systems with RAG to accelerate mainframe migration and data pipeline development. Built comprehensive solutions with responsive frontends and scalable backend APIs on PySpark and Databricks. Scaled a vectorstore for a real-time call center chatbot copilot.

- **Project: genAI System with RAG for Accelerating Data Pipeline Development**
 - Implemented FastAPI API to expose source to target data transformation as a REST service.
 - Designed and implemented parallel calls to openAI API by using asyncio to improve performance.
 - Designed and documented the local development and debug workflow using debugpy with VSC and PyCharm.
 - Prompts tuning to generate data pipeline PySpark function code and unit test code.
- **Technologies:** Python/Flask/FastAPI/Pydantic, TypeScript/JavaScript/NodeJs, genAI/openAI/Langchain, Microservices, REST/JSON, Databricks/pyspark, Event Driven Architecture (EDA), Docker/K8s/EKS

Staff Engineer, Yotascale

Dec., 2020 – Dec., 2021, Cherry Hill, NJ (WFH), USA

Led the architecture and full-stack development of a cloud cost analytics platform. Introduced a Microservice Design First methodology based on OpenAPI/Swagger, ensuring clean, consistent, and well-documented APIs across the product.

- **Project: Lead Architect for Cloud Cost Analytics Platform**
 - Introduced and built Microservice Design First Methodology and team culture based on OpenAPI/Swagger Specification.
 - Designed and developed Cost Analytics Microservices using SpringBoot with ClickHouse and PostgreSQL.
 - Designed and implemented dynamic metadata driven widget/filter configuration Microservice.
 - Implemented Junit test cases with local docker test containers.

- **Technologies:** OpenAPI/Swagger, Microservice, Java/Kotlin/SpringBoot, Python/Flask, Kubernetes/EKS, Terraform, REST API Design, JSON Schema

Lead Software Engineer, Oracle

May, 2014 – June, 2017, Redwood, CA (WFH@NJ), USA

Developed SDKs and integration frameworks for Oracle's Cloud and SOA Suite, focusing on API design, web services, and creating tools for developers.

- **Project: Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework**
 - Designed and implemented new APIs for the SDK to provide cache and pagination abilities.
 - Programmatically generated and regenerated integration WSDL from source files.
 - Troubleshoot WSDL/XSD generation issues related to complex mixing namespaces and prefixes.
- **Technologies:** Java, SOA, Web Services, SOAP/WSDL, REST/JSON/XML, API, SDK, Docker

Education

Master of Computer Science, Texas Tech University

Sept., 2002 – Jun., 2006, Lubbock, TX, USA

GPA: 4.0/4.0

Master of Engineering, Shanghai JiaoTong University

Sept., 1993 – Feb., 1996, Shanghai, China

Major: Hydraulic Fluid Control in Mechanical Engineering

Bachelor of Engineering, Changsha Railway University

Sept., 1987 – Jul., 1991, Changsha, Hunan, China

Major: Engineering Machine

Languages

- English: Fluent
- Chinese: Native

Hobbies

- iOS/Android Apps Development: Developing a Chinese character learning App
- Playing with cutting-edge technologies: genAI/Autogen/Multiagents, JavaScript, Node.js, React
- **Project: Personalized JSON Schema Resume**
 - Adapted the JSON schema and developed new data filtering utility methods in JavaScript.
 - Personalized CSS3 styles for responsive design and print formatting.

- Generated PDF files from HTML using Puppeteer.
- **Technologies:** JavaScript, HTML5, CSS3, Node.js